Corporate Bankruptcy Risk Prediction with Machine Learning

Comparing Interpretable Logistic Regression
and Tree-Based Models

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1 Introduction

1.1 Financial motivation

Corporate bankruptcy is an important problem for lenders, investors, auditors, and risk managers. A firm's failure can impose losses on creditors and shareholders. It can also disrupt relationships with employees, suppliers, and other firms. Financial statements provide structured information about assets, liabilities, revenues, expenses, and profitability. These variables may contain signals of financial distress, so statistical learning can be used to study bankruptcy risk. Early research showed that accounting variables can help distinguish distressed from healthy firms (Altman, 1968). Later work studied bankruptcy risk in a probabilistic classification framework (Ohlson, 1980).

Despite this long research history, bankruptcy classification remains difficult. The first difficulty is class imbalance. Failure is rare, so a model can achieve high overall accuracy by predicting that almost every observation is alive. Such a model may look successful even though it detects none of the failed observations. A second difficulty is the structure of financial statement data. Financial variables are often correlated, measured on different scales, and partly affected by firm size. The relationship between accounting variables and failure may also be nonlinear or depend on interactions. A third difficulty is economic interpretation. Missing a failed firm can be costly, but flagging too many healthy firms can make a screening system impractical.

For these reasons, evaluation is as important as model fitting. Accuracy is reported, but it is not the main criterion in this imbalanced setting. The paper therefore emphasizes failed-class precision, recall, F1-score, balanced accuracy, ROC-AUC, and average precision, reported as PR-AUC. Precision–recall analysis is especially useful when the negative class is much larger than the positive class, because it shows the false-positive burden more directly (Davis and Goadrich, 2006). The classification threshold is also treated as a decision choice rather than as an automatically optimal value.

1.2 Research question and contribution

This paper investigates the following research question:

> *How effectively can interpretable and tree-based machine-learning models distinguish failed from alive company-year observations for previously unseen companies, and what trade-offs arise between failure detection and false alarms?*

The analysis uses the American corporate bankruptcy dataset introduced by Lombardo et al. (2022). The working sample contains 78,682 company-year observations from 8,971 anonymized companies between 1999 and 2018, with 18 financial predictors and a binary failed/alive target. Because companies may appear in multiple years, observations are split at the company level. This prevents the same company from appearing in both training and final-test data and reduces the risk that persistent company characteristics create an artificially favorable evaluation. The design tests generalization to unseen companies; it is not presented as a chronological forecasting backtest.

The empirical comparison begins with a majority-class baseline, which demonstrates why accuracy alone is misleading. Logistic Regression provides the main interpretable benchmark. L1- and L2-regularized variants test whether coefficient shrinkage helps when predictors are correlated. A Decision Tree, Random Forest, and Gradient Boosting model then test whether nonlinear rules and interactions improve performance. Together, these models allow a comparison between a simple interpretable model and more flexible tree-based methods. This makes it possible to compare models that differ in flexibility and interpretability (James et al., 2023). A PCA plus Logistic Regression specification is included as a dimension-reduction extension, not as the main model.

The paper focuses on three main points. First, it compares interpretable and tree-based models in a reproducible way and avoids company overlap between the training and test data. Second, it uses metrics that focus on the failed class and keeps the final test set separate for the main comparison. Third, it connects the model results to financial screening by discussing classification errors, threshold trade-offs, coefficients, and feature importance. The results are interpreted as predictive associations rather than causal effects. The project is intended as an educational model comparison and not as a system for automatic credit or investment decisions.

2 Data

2.1 Dataset and unit of observation

The project uses the American corporate bankruptcy dataset published by Lombardo et al. (2022). It contains accounting information for public companies listed in the United States. The version used here has 78,682 company-year observations covering the years 1999 to 2018. It contains 8,971 anonymized company identifiers, 18 financial predictors, a year variable, and a company status. The data summary reports no missing values and no duplicated rows.

One row represents one company in one year. The same company can therefore appear in several years. This matters for the train-test design. A random row-level split could place observations from the same company in both training and test data. The model could then benefit from company-specific information already seen during training. To reduce this risk, the project assigns whole companies to the training or final-test sample. Section 3 explains the split in more detail.

The source data cover a long period in which both the set of listed companies and the wider economic environment changed. The project does not use a time-based holdout. The final test set contains unseen companies, but its calendar years overlap with the training years. The results therefore show how well the models work for unseen companies under this split, not how well they would predict a later period.

2.2 Target and financial predictors

The original status is recorded as `alive` or `failed`. For modeling, failed observations are coded as 1 and alive observations as 0. The failed class is treated as the positive class throughout the analysis. The company identifier and year are kept for splitting and reporting, but they are not used as predictors. This leaves 18 financial inputs, labelled `X1` to `X18` in the source file.

The predictors cover several parts of a company's financial position and performance. They include current assets, inventory, receivables, market value, and total assets; profitability measures such as EBITDA, EBIT, net income, gross profit, and retained earnings; long-term debt, current liabilities, and total liabilities; and activity measures such as sales, revenue, cost of goods sold, and operating expenses. Depreciation and amortization is also included. The models therefore use the reported financial values rather than newly created financial ratios. The tables and figures use readable names so that the outputs can be discussed in financial terms.

The variables have very different scales. Several also describe related parts of the same financial statements. For example, revenue and net sales are closely connected, as are assets and liabilities. This matters for Logistic Regression because coefficient size depends on measurement scale. Correlated predictors can also make individual coefficients less stable. The Logistic Regression and PCA pipelines therefore standardize the predictors using parameters learned from the training data. Tree-based models do not require this scaling.

2.3 Class imbalance and descriptive evidence

The target is strongly imbalanced. Of the 78,682 company-year observations, 73,462 are alive and 5,220 are failed. The failed class therefore represents only 6.63% of the full sample. Figure 1 shows this difference. A classifier that predicts every observation as alive would be correct in more than 93% of the full sample. However, it would detect no failures. This is the main reason why the later evaluation does not rely on accuracy alone.

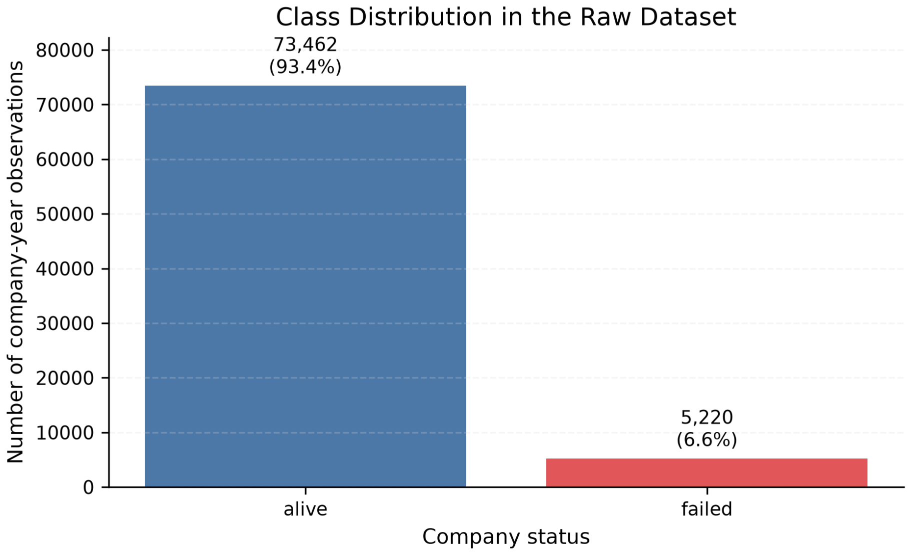

Figure 1: Class distribution in the full dataset. The failed class accounts for 6.6% of company-year observations.

The share of failed observations also changes over time. As shown in Figure 2, the observed rate rises from 7.16% in 1999 to a peak of 9.40% in 2003 and then generally declines, reaching 1.32% in 2018. This pattern should be interpreted carefully. It may reflect economic conditions, changes in the companies included in the dataset, or the way status is represented. The figure only describes the available sample. It does not show that time caused the change in failure rates.

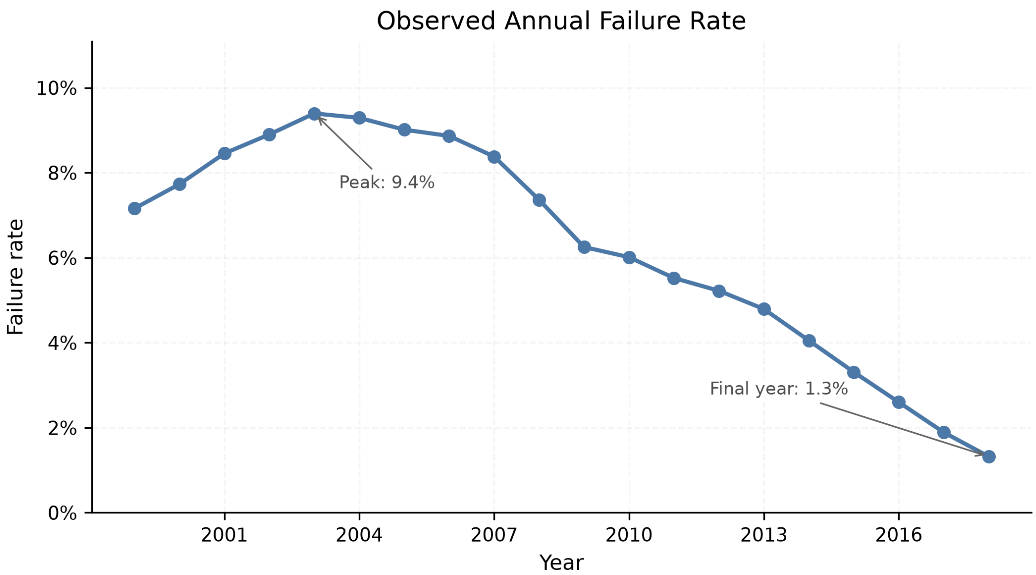

Figure 2: Observed annual failure rate in the full dataset, 1999–2018.

Table 1 compares the medians of six selected financial variables. Medians are used because financial statement values are often skewed and can contain very large observations. Failed observations have lower median market value, net income, current assets, and retained earnings. They have higher median long-term debt and total liabilities. These differences make sense from a financial perspective, but they do not show that the variables cause failure. They are simple comparisons between the two groups and do not control for the other predictors.

Table 1: Selected financial-variable medians by company status

Financial variable	Alive median	Failed median
Market value	240.90	117.80
Net income	2.07	-3.33
Total long-term debt	7.09	18.44
Current assets	102.92	75.87
Total liabilities	80.74	97.04
Retained earnings	-0.17	-25.51

Notes: Values are shown in the units reported by the source dataset. The table is descriptive and does not imply causal effects.

3 Methodology

3.1 Company-level training, validation, and test design

The full sample is first divided into training and final-test data with a grouped random split. The company identifier is used as the group, so every row from a company is assigned to the same side of the split. The split places 80% of companies in training and 20% in the final test set. A fixed random seed of 42 is used so that the result can be reproduced. The training set contains 62,988 rows from 7,176 companies, while the final test set contains 15,694 rows from 1,795 companies. The failure rates are 6.42% and 7.50%, respectively.

A second company-level split is then created inside the training set. This inner split reserves 20% of the training companies for validation. It produces a model-fitting sample of 50,323 rows from 5,740 companies and a validation sample of 12,665 rows from 1,436 companies. Hyperparameters are compared on the validation sample using PR-AUC. The same validation predictions are later used for the threshold analysis. The final test set is not used for model selection or validation threshold analysis.

The final evaluation uses the fitted models from the inner model-fitting sample. The validation rows are not added back for a final refit. This means that the final models are fitted with fewer observations than they could have used. It is also less data-efficient than refitting the selected specification on all non-test training rows. That alternative was not used here, and this point is considered again in the limitations section.

All random model components use the same seed. The implementation uses scikit-learn pipelines and estimators (Pedregosa et al., 2011). Pipelines are important for Logistic Regression and PCA because standardization is learned only from the fitting data and then applied to validation or test observations. This prevents information from the validation or test data from entering the preprocessing step.

3.2 Majority-class baseline

The first model is a majority-class baseline. It always predicts the most common training label, which is `alive`. The baseline does not try to learn a relationship between the financial variables and the target. Its purpose is to give a simple reference point. Because alive observations are very common, this rule achieves high accuracy while its recall for failures is zero. A useful bankruptcy classifier should therefore be evaluated with more than overall accuracy.

3.3 Logistic Regression and regularization

Logistic Regression is the main interpretable model. For observation i, the model links the probability of failure p_i to the financial predictors through

$$\log\left(\frac{p_i}{1-p_i}\right) = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_{18} x_{i18}. \tag{1}$$

The left-hand side is the log-odds of failure. A positive coefficient is linked to higher predicted failure log-odds when the other variables are held fixed. A negative coefficient is linked to lower predicted failure log-odds. The coefficients are not direct percentage-point changes in probability, and they are not interpreted as causal effects.

Each Logistic Regression specification is placed after a standardization step. The predictors therefore have mean zero and standard deviation one in the model-fitting sample. This makes the regularization penalty comparable across variables with different units. Balanced class weights give more importance to errors on the minority failed class during fitting. They do not create additional failed observations, rebalance the test set, directly choose the classification threshold, or make the resulting scores calibrated real-world bankruptcy probabilities.

Three Logistic Regression specifications are compared. The main benchmark uses an L2 penalty with fixed $C = 1$. L1 and L2 variants are added to check whether regularization helps when predictors are correlated. The parameter C is the inverse of the regularization strength, so a smaller value means stronger shrinkage. Separate L1 and L2 models compare $C \in \{0.01, 0.1, 1, 10\}$ using validation PR-AUC. Both regularized models select $C = 0.1$. L1 regularization can set some coefficients exactly to zero. L2 regularization usually keeps all variables but shrinks their coefficients. These methods can help when predictors are strongly related, although they can also add some bias (James et al., 2023).

3.4 Tree-based models

Tree-based models are included because they can represent nonlinear patterns and interactions without manually specifying them. The Decision Tree is the simplest tree model in the comparison. It separates the observations by applying a sequence of decision rules. Its depth and minimum leaf size control complexity. A deep tree may fit the training data closely but can have high variance. The selected tree has a maximum depth of 5 and at least 50 observations in each leaf. Balanced class weights are used during fitting.

Random Forest is included to reduce the variance of a single tree. It averages many trees fitted on resampled data. It also considers a random subset of predictors at each split, which makes the trees less similar to each other (Breiman, 2001). The selected model uses 300 trees, no fixed maximum depth, at least 50 observations per leaf, and the square root of the available predictors at each split. Class weights are balanced separately within each bootstrap sample.

Gradient Boosting is included as a second ensemble approach. It also combines many trees, but it builds them in sequence. Each new tree is added to improve the errors left by the existing ensemble (Friedman, 2001). The learning rate controls how much each new tree changes the model. The selected specification uses 150 trees, a learning rate of 0.05, a maximum tree depth of 3, and at least 100 observations per leaf. The scikit-learn Gradient Boosting estimator does not have a direct class-weight argument, so balanced observation weights are supplied during fitting.

The parameter grids are kept small. They include the main settings that control model complexity without requiring a very large search. The complete notebook workflow and saved output tables document the exact selected specifications.

3.5 PCA extension

Principal Component Analysis (PCA) is included as a separate course extension. PCA replaces the original standardized predictors with linear combinations called principal components. The first component captures the largest possible share of variation in the predictors. Each later component captures another orthogonal direction of variation (Jolliffe and Cadima, 2016). This can summarize correlated financial variables with fewer inputs.

The PCA pipeline first standardizes the 18 predictors, then applies PCA, and finally fits a balanced Logistic Regression with $C = 1$. The tested component counts are 2, 3, 5, 8, 10, 12, 15, and 18. They are compared on the same type of internal company-level validation split used for the main models. The PCA results are validation results only. PCA is not used as a final-test model-selection stage.

PCA can reduce the number of inputs, but it also has two disadvantages in this project. First, it is unsupervised: components are chosen to explain variation in the predictors, not to separate failed and alive observations. A component can explain a large share of the predictor variation without being useful for classification. Second, each component mixes several financial variables. This makes the model harder to explain than Logistic Regression fitted directly on current assets, income, debt, and the other original variables.

4 Evaluation Strategy

4.1 Why accuracy is insufficient

Accuracy is the share of all observations classified correctly. Although it is easy to understand, it can be misleading when one class is much more common than the other. In this dataset, a model can obtain high accuracy by predicting almost every observation as alive. Such a model would still fail at the main task because it would detect no failures. The majority-class baseline is included to make this problem visible.

The confusion matrix shows the different types of errors more clearly. Failed firms are treated as the positive class. A true positive is a failed observation that the model detects. A false negative is a failed observation that the model misses. A false positive is an alive observation flagged as failed. A true negative is an alive observation correctly classified as alive. In a screening setting, false negatives are missed failures and false positives are false alarms. Both types of error matter, but they may not have the same cost.

Balanced accuracy is also reported. It is the average of the recall for the failed class and the recall for the alive class:

$$\text{Balanced accuracy} = \frac{1}{2}\left(\frac{TP}{TP+FN} + \frac{TN}{TN+FP}\right). \tag{2}$$

Unlike ordinary accuracy, this measure gives equal weight to both classes.

4.2 Ranking and failed-class metrics

The evaluation gives special attention to the minority failed class. Failed precision measures how many observations predicted as failed actually failed:

$$\text{Precision}_{\text{failed}} = \frac{TP}{TP+FP}. \tag{3}$$

Failed recall measures how many actual failures were detected:

$$\text{Recall}_{\text{failed}} = \frac{TP}{TP+FN}. \tag{4}$$

Their harmonic mean is the failed-class F1-score:

$$F1_{\text{failed}} = 2\frac{\text{Precision}_{\text{failed}} \times \text{Recall}_{\text{failed}}}{\text{Precision}_{\text{failed}} + \text{Recall}_{\text{failed}}}. \tag{5}$$

Recall is important when missed failures are a concern, while precision shows how many false alarms are produced. The F1-score combines precision and recall, but it does not show the financial cost of either type of error.

Two threshold-independent ranking measures are also reported. ROC-AUC measures how well the model generally ranks failed observations above alive observations across different thresholds. However, ROC curves can look favourable when the negative class is much larger. Precision–recall analysis focuses more directly on the positive minority class and is therefore especially useful here (Davis and Goadrich, 2006).

In the project output, the measure labelled PR-AUC is calculated using the average precision implementation in scikit-learn (Pedregosa et al., 2011). Average precision summarizes how precision changes as recall increases. This distinction is mentioned because average precision is not exactly the same as the trapezoidal area under a precision–recall curve.

The ranking measures use predicted scores across many possible cut-offs. In contrast, accuracy, precision, recall, F1, and the confusion matrix require one classification threshold and a set of predicted labels. Ranking metrics ask how well the model orders observations. Threshold-based metrics describe the classifications made at one cut-off.

4.3 Threshold analysis

The default classification threshold of 0.50 is not necessarily the most appropriate choice for bankruptcy screening. Lowering the threshold usually detects more failed firms, but it also creates more false alarms. Raising it usually improves precision at the cost of lower recall. The threshold should therefore depend on how the model will be used.

Thresholds from 0.01 to 0.99 are examined using only the internal validation predictions for Logistic Regression, Random Forest, and Gradient Boosting. The analysis considers three rules: the threshold with the highest failed-class F1-score, the highest-precision threshold that still reaches at least 60% failed recall, and the corresponding rule for at least 80% failed recall. These rules show what happens when failure detection receives more weight.

The final test set is not used to choose a threshold. To keep the main model comparison consistent, the reported final-test classifications use each estimator's default decision rule, equivalent to a 0.50 probability cut-off for the models considered. The validation threshold analysis is used only to show how the results change and is not treated as a final decision rule. A real decision rule would also require calibrated probabilities, explicit financial costs, and evidence from a later or external sample. These elements are outside the scope of this project.

5 Results

5.1 Final-test model comparison

Figure 3 compares the majority-class baseline with Logistic Regression, Random Forest, and Gradient Boosting on the held-out test set. These models are shown because they represent the main differences in model performance. The complete saved results also contain the regularized logistic models and the single Decision Tree. Gradient Boosting is the model selected by validation PR-AUC, while the final test set is used only for evaluation.

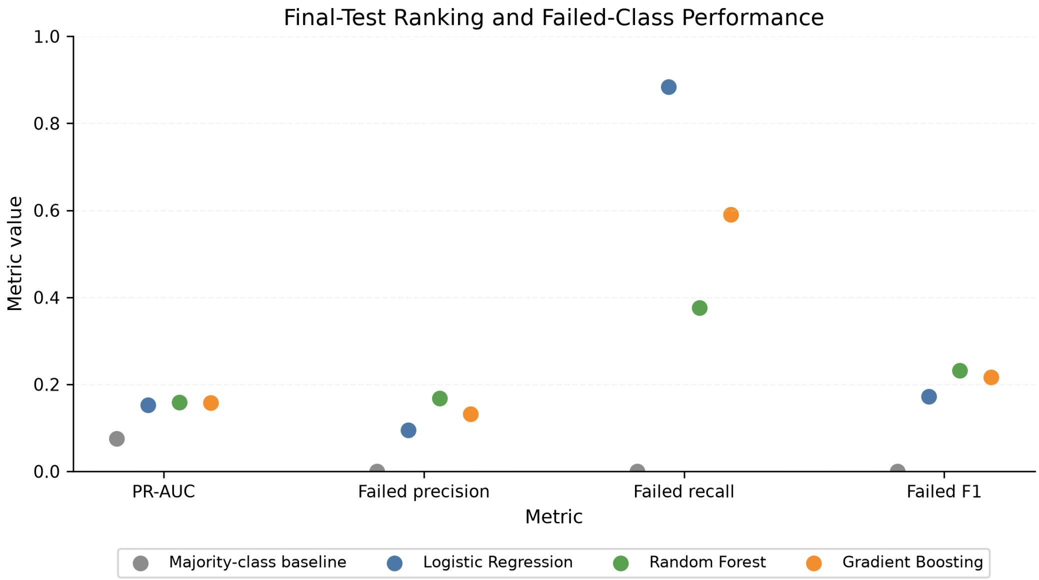

Figure 3: Final-test performance for the failed class. PR-AUC refers to average precision. The values come from the fixed held-out test set.

The majority baseline reaches 0.925 accuracy, but it detects none of the 1,177 failed observations. Its failed-class precision, recall, and F1-score are all zero. This confirms that accuracy alone gives the wrong impression in this imbalanced problem.

No model is strongest on every measure. Logistic Regression has the highest failed recall at 0.884, but its failed precision is only 0.095. Random Forest has the numerically highest observed final-test PR-AUC among the models shown at 0.159 and the highest failed-class F1-score at 0.232. This result was observed on the final test set and does not change the earlier model-selection decision. Gradient Boosting has the highest balanced accuracy at 0.638 and an intermediate recall of 0.590. The ROC-AUC values of the three models are close, ranging from 0.688 to 0.699. Overall, the models perform differently depending on the metric being considered.

5.2 Classification trade-offs

Table 2 shows the final-test predictions as practical screening results. Logistic Regression detects 1,041 failures and misses only 136. However, it also creates 9,906 false alarms. This explains why its recall is high while its precision and overall accuracy are low. Balanced class weighting helps explain this result. Failed observations receive more influence during fitting, so the fitted model screens broadly at the default decision rule. The 0.50 rule is a conventional comparison point rather than an economically optimized threshold.

Table 2: Final-test classification outcomes at the default decision rule

Model	Correct alive	False alarms	Missed failures	Detected failures
Majority baseline	14,517	0	1,177	0
Logistic Regression	4,611	9,906	136	1,041
Random Forest	12,327	2,190	735	442
Gradient Boosting	9,954	4,563	482	695

Random Forest is more selective. It reduces false alarms to 2,190, but detects only 442 failures and misses 735. Gradient Boosting lies between these two patterns, with 695 detected failures and 4,563 false alarms. In practical terms, Logistic Regression flags the most companies, Random Forest gives fewer alerts, and Gradient Boosting lies between them. Which pattern is preferable depends on the relative cost of a missed failure and a false alarm. Those costs are not estimated in this project.

The validation threshold analysis shows the same trade-off. For each model, validation-based scenarios that reach at least 80% recall label a much larger share of observations as failed than the maximum-F1 scenario. Recall can therefore be increased, but this comes with lower precision and more firms sent for further review. These cut-offs were studied on validation data only. They are not used to revise the final-test results above.

5.3 Precision–recall analysis

Figure 4 shows the precision–recall curves for the three key models. The dashed horizontal line represents the failed-class share in the test data. A curve above this line shows better ranking than a non-informative classifier at that recall level.

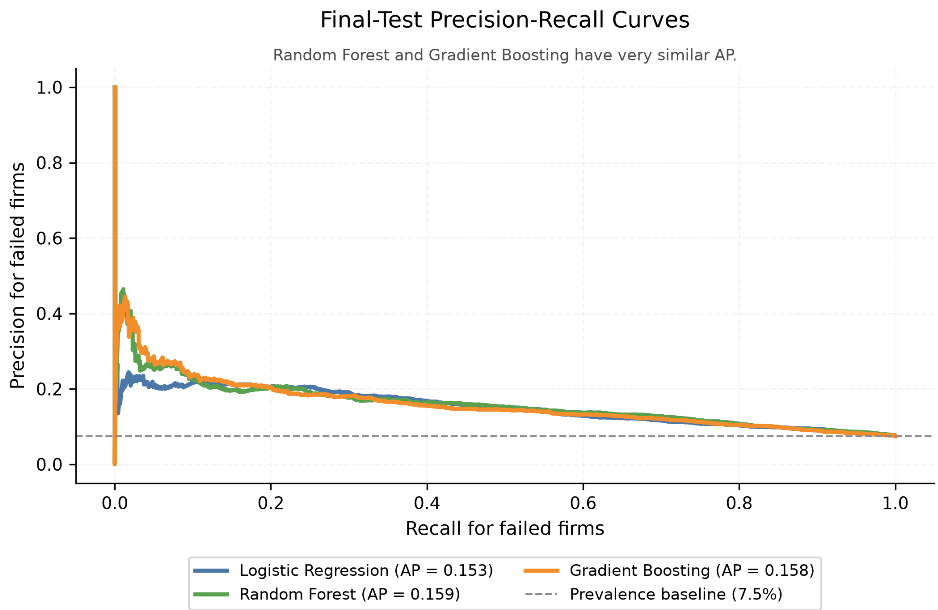

Figure 4: Final-test precision–recall curves for the key models. The legend reports average precision, labelled PR-AUC in the project outputs.

The curves are close. Their average precision values are 0.153, 0.159, and 0.158. Random Forest is numerically highest, but its advantage over the validation-selected Gradient Boosting model is very small. Logistic Regression has similar ranking performance, but its default-threshold classifications produce a very different confusion matrix. This shows why ranking quality and threshold-based decisions must be discussed separately. Overall, the models contain useful predictive information, but the low precision values show that identifying failed companies remains difficult.

6 Model Interpretation

6.1 Logistic Regression coefficients

Figure 5 shows the largest coefficients from the main Logistic Regression. Because the predictors were standardized before fitting, the coefficient sizes can be compared more directly. Positive coefficients are associated with higher predicted failure log-odds. Negative coefficients are associated with lower predicted failure log-odds. These relationships are interpreted while the other predictors are kept constant.

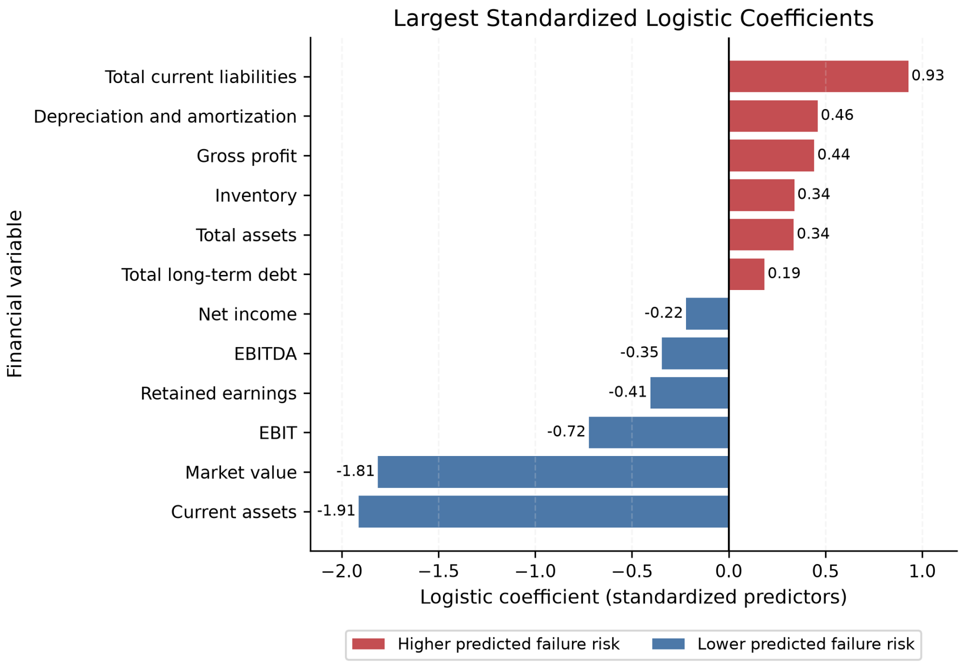

Figure 5: Largest Logistic Regression coefficients. Predictors are standardized, and the target value one represents a failed observation.

Current assets and market value have the largest negative coefficients, at -1.912 and -1.812. EBIT, retained earnings, and EBITDA also have negative coefficients among the variables shown. In this fitted model, higher values of these variables are associated with lower predicted failure risk. This is also similar to the descriptive results, where failed observations have lower median current assets, market value, net income, and retained earnings.

Total current liabilities have the largest positive coefficient at 0.928. Depreciation and amortization, gross profit, inventory, and total assets also have positive coefficients among the largest values. These coefficient signs should be interpreted carefully. The model includes several related accounting totals at the same time. A coefficient therefore describes a conditional relationship after the other variables are held fixed. For example, the positive gross-profit coefficient should not be read as evidence that higher gross profit causes bankruptcy. The positive sign may result from the way gross profit is related to company size, revenue, costs, and balance-sheet variables in the fitted model.

The coefficients help explain how the model forms its scores, but they are not causal estimates. The analysis also does not report statistical inference for individual coefficients. They are used here only to explain the model's predictions.

6.2 Tree-based feature importance

Figure 6 reports the leading impurity-based feature importance values for Random Forest and Gradient Boosting. A larger value means that the variable contributed more to reducing impurity when the model created splits. Unlike a logistic coefficient, this measure does not show whether higher values increase or decrease predicted risk.

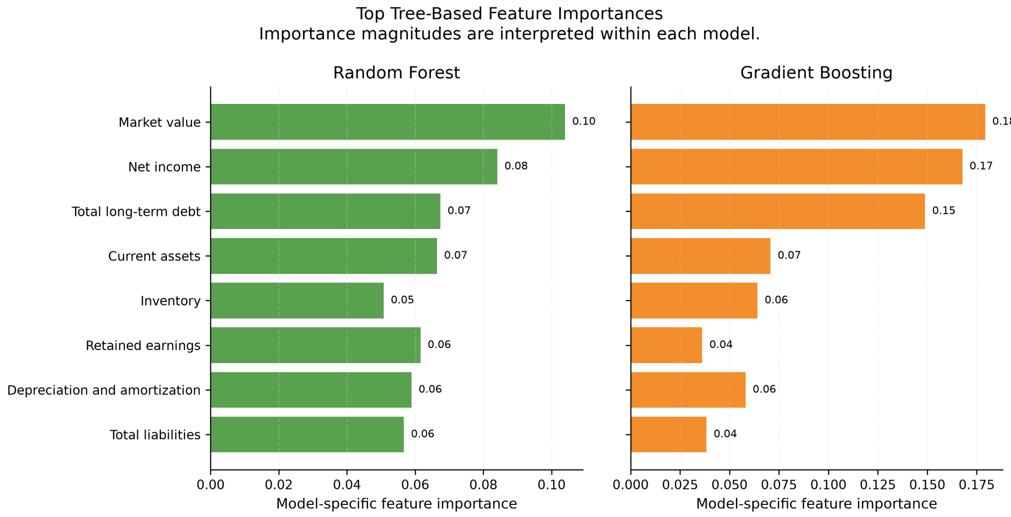

Figure 6: Leading feature importance values within the Random Forest and Gradient Boosting models. Magnitudes should be compared mainly within each model rather than directly across models.

Market value, net income, and total long-term debt are the three most important features in both ensembles. Current assets also appear near the top, followed by variables such as inventory, depreciation and amortization, and total liabilities. Retained earnings is also prominent in the Random Forest. This overlap suggests that market valuation, profitability, liquidity, and debt information all contribute to the models' predictions. It does not establish that these variables cause failure.

The two models also distribute importance differently across the variables. Random Forest spreads importance more evenly across the predictors, while Gradient Boosting assigns a larger share to its first few variables. The absolute values should not be compared as if they were on one common scale. Each ensemble constructs and combines trees differently. Impurity-based importance can also favour variables that provide many useful split points. When predictors are correlated, importance can be divided across related variables. The figure should therefore be used to describe the fitted models, not to rank the economic causes of bankruptcy.

Taken together, the two interpretation methods offer different information. Logistic Regression provides signed and relatively direct relationships on the log-odds scale. The ensembles allow nonlinear patterns and interactions, but their importance values are less direct. This shows the main trade-off between model flexibility and interpretability in the project. More flexible models can represent richer patterns, while the logistic model is easier to explain in financial terms.

7 PCA Results

The PCA analysis examines whether the 18 standardized financial variables can be reduced without losing much predictive information. The first few components explain a large share of predictor variation. Two components retain 83.8% of total variance, five retain 93.9%, and ten retain 99.2%.

A high explained variance does not automatically mean strong classification performance. Table 3 reports the internal validation results for PCA followed by Logistic Regression. With only two components, ROC-AUC is 0.498 and PR-AUC is 0.063, even though those components retain most of the predictor variance. Performance improves when more components are included. The ten-component version has the highest failed-class F1-score, at 0.152. The twelve-component version has the highest PR-AUC, at 0.151. These differences are small.

Table 3: Internal validation results for PCA with Logistic Regression

Components	Explained variance	ROC-AUC	PR-AUC	Failed F1
2	0.838	0.498	0.063	0.130
3	0.892	0.642	0.116	0.146
5	0.939	0.651	0.124	0.148
8	0.980	0.669	0.140	0.150
10	0.992	0.671	0.147	0.152
12	0.998	0.674	0.151	0.150
15	1.000	0.671	0.147	0.149
18	1.000	0.671	0.147	0.149

For comparison, Logistic Regression on the original standardized predictors has a validation PR-AUC of 0.148 and a failed-class F1-score of 0.149. The PCA models with ten or twelve components therefore perform similarly, but they are not clearly better. PCA can reduce the input dimension. However, reducing the data to only two or three components removes information that appears useful for distinguishing failed and alive observations.

PCA also makes the results harder to interpret in financial terms. Each component is a weighted combination of many accounting variables, so it cannot be discussed as directly as current assets, market value, income, or debt. In this project, the small improvement on the validation sample is not enough to justify the weaker interpretation. The PCA analysis is still useful because it shows that explaining predictor variation is not the same as predicting bankruptcy well. All PCA comparisons are validation results only. They are not used to make any additional conclusions about final-test performance.

8 Critical Reflection and Limitations

The project includes several choices that strengthen the analysis. Splitting by company prevents the same firm from appearing in both training and test data. Keeping the test set separate from model selection and threshold analysis also makes the final comparison more reliable. Finally, the majority baseline and failed-class metrics expose weaknesses that would be hidden by accuracy. These choices improve the analysis, but they do not remove the following limitations.

8.1 Data and target limitations

The data contain repeated company-year observations. Observations from one firm are likely to be related over time. The company-level split handles direct firm overlap, but the design is not a chronological forecasting exercise. The training and test samples cover overlapping calendar years. The results therefore show how well the models work for unseen companies in a similar historical period, not how well they would perform in a later period. The changing annual failure rate also suggests that time and sample composition matter.

The failed/alive label is suitable for classification, but the future period being predicted is not clearly defined. A real early-warning application would need a statement such as “failure within the next twelve months” and would need to confirm that every accounting variable was available before that forecast date. Without this timing detail, the results should be read as classification of company-year observations, not as a complete real-time forecasting exercise.

The predictors are financial statement levels. They contain information about firm condition, but they may also partly reflect company size and accounting scale. This matters because large firms naturally have larger assets, liabilities, sales, and expenses. The project does not add financial ratios, changes over time, industry indicators, macroeconomic conditions, governance information, or market returns. Dataset coverage and reporting choices may also make the sample different from other countries, private firms, or later periods. For this reason, the results may not apply in the same way to other datasets or settings.

8.2 Validation and modelling limitations

The reported results come from one grouped train–test split and one internal validation split. They do not show how much performance would vary under other valid splits. There are no confidence intervals, repeated grouped validation, or statistical tests of the small differences between models. In particular, the very close PR-AUC values do not prove that one ensemble model is generally better than the other.

The hyperparameter searches are limited so that the project remains manageable, but better settings may still exist. The saved final models are also fitted on the inner model-training portion and are not refitted after validation. This means that some available training observations are not used in the final model. It is also less data-efficient. A later version could lock all choices, refit on the complete non-test sample, and then evaluate once on a still-unused test set.

The models evaluate discrimination and classification, not probability calibration. Their outputs should therefore not automatically be interpreted as literal bankruptcy probabilities. Interpretation also has limits. Logistic coefficients can change when correlated predictors are included together. Impurity-based tree importance has its own biases. Both are descriptions of fitted predictive models, not causal evidence.

8.3 Decision and practical limitations

The threshold analysis makes the precision–recall trade-off visible, but the project does not estimate the monetary cost of a missed failure or a false alarm. It also does not specify how many flagged firms an analyst could review. Without clear financial costs or a limit on how many cases can be reviewed, it is not possible to identify one best model or threshold. The low failed-class precision is especially important: even the more selective models produce many false alerts.

A real financial application would require further checks, including probability calibration, chronological backtesting, stability across industries and economic conditions, stress testing, fairness and governance review, and a plan for monitoring model drift. Human review would also be necessary before any credit, investment, or regulatory action. The current project is a reproducible educational study and screening exercise. It is not production-ready.

8.4 Priorities for future work

The first improvement would be to define a clear forward failure horizon and use rolling or chronological evaluation. Repeated company-grouped validation and uncertainty intervals would show whether model differences are stable. Financial ratios and trends could reduce the influence of firm size, while industry and macroeconomic variables could add context. After model selection, probability calibration should be tested, and the threshold should be chosen using explicit financial costs and review capacity. These improvements would be more useful than simply adding another complex model.

9 Conclusion

This paper examined whether financial statement variables can help identify failed observations for companies that were not used during model fitting. It compared interpretable and tree-based classifiers for company-year bankruptcy-risk classification. The results show that the financial variables contain useful information for distinguishing failed and alive observations under the company-level split. However, bankruptcy classification remains difficult, and the best model depends on the goal of the analysis.

The majority-class baseline achieved 92.5% accuracy while detecting no failed observations. This shows why accuracy alone is insufficient. Among the key models, Logistic Regression had the highest failed recall, but it also produced many false positives. Random Forest produced the numerically highest observed final-test PR-AUC and failed-class F1-score. Gradient Boosting was the validation-selected model and had the highest balanced accuracy. This is important because the final test set is used only for evaluation and not to select the model afterward.

The additional analyses show the same general pattern. Validation threshold analysis showed that higher recall requires lower precision and more review effort. PCA reduced the number of inputs but did not clearly improve validation performance. Logistic Regression coefficients and tree-based feature importance helped explain the fitted models, especially for valuation, profitability, liquidity, and debt variables, but these relationships are not causal. Overall, the models are useful for screening companies in this course project, but they should not be used for automatic financial decisions or as a production system. A future study should define a clear prediction period, use chronological evaluation, test calibration and uncertainty, and choose thresholds based on financial costs and review capacity.

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